



Persistent Visual and Vestibular Impairments for Postural Control Following Concussion: A Cross-Sectional Study in University Students

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Abstract

Objective To examine how concussion may impair sensory processing for control of upright stance.

Methods Participants were recruited from a single university into 3 groups: 13 participants (8 women, 21 ± 3 years) between 2 weeks and 6 months post-injury who initiated a return-to-play progression (under physician management) by the time of testing (recent concussion group), 12 participants (7 women, 21 ± 1 years) with a history of concussion (concussion history group, > 1 year post-injury), and 26 participants (8 women, 22 ± 3 years) with no concussion history (control group). We assessed sensory reweighting by simultaneously perturbing participants' visual, vestibular, and proprioceptive systems and computed center of mass gain relative to each modality. The visual stimulus was a sinusoidal translation of the visual scene at 0.2 Hz, the vestibular stimulus was ± 1 mA binaural monopolar galvanic vestibular stimulation (GVS) at 0.36 Hz, the proprioceptive stimulus was Achilles' tendon vibration at 0.28 Hz.

Results The recent concussion (95% confidence interval 0.078–0.115, $p=0.001$) and the concussion history (95% confidence interval 0.056–0.094, $p=0.038$) groups had higher gains to the vestibular stimulus than the control group (95% confidence interval 0.040–0.066). The recent concussion (95% confidence interval 0.795–1.159, $p=0.002$) and the concussion history (95% confidence interval 0.633–1.012, $p=0.018$) groups had higher gains to the visual stimulus than the control group (95% confidence interval 0.494–0.752). There were no group differences in gains to the proprioceptive stimulus or in sensory reweighting.

Conclusion Following concussion, participants responded more strongly to visual and vestibular stimuli during upright stance, suggesting they may have abnormal dependence on visual and vestibular feedback. These findings may indicate an area for targeted rehabilitation interventions.

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Key Points

Young adults with a recent concussion or a history of concussion appear to have higher gains to visual and vestibular stimuli than controls, suggesting abnormal dependence on visual and vestibular feedback in destabilizing environments.

Young adults with a recent concussion have worse conventional sway measures than those with a concussion history and controls, suggesting noisier sensory feedback among those with a recent concussion, for which those with a more distant concussion history restored function or learned to compensate through greater reliance on sensory integration.

Neither young adults with a recent concussion nor those with a concussion history appear to have impairments in sensory reweighting relative to controls.

1 Introduction

Although concussion symptoms typically resolve within 2–4 weeks [1, 2], neurophysiological impairments may persist beyond symptom recovery [3]. For example, several studies have suggested that postural control impairments last longer than 2–4 weeks post-injury [4–6]. Such impairments may be detectable by measuring movement of the center of mass (COM) and/or the center of pressure (COP) using sophisticated laboratory equipment [7–9]. Without these resources, clinicians score clinical assessments, such as the Balance Error Scoring System (BESS), Romberg, or tandem gait tests, by visual inspection and timing (e.g., pass/fail over 20–30 s) [7–10]. Clinical assessments indicate postural control impairments recover within 2–5 days post-injury, but quantitative assessments indicate persistent impairments beyond clinical recovery [4–6]. Physicians often allow athletes to return to play as soon as they have been cleared on clinical measures. Residual physiological deficits raise the concern of whether some return-to-play decisions are premature, which may have consequences. For example, recent research suggests that athletes are at 2–3 times increased risk for subsequent concussion and/or subsequent musculoskeletal injury following recovery from concussion [2, 11–15]. This may be due to maladaptive sensorimotor control after concussion [14, 16–19], although the specific physiological mechanisms underlying postural control impairments following concussion are unknown.

The central nervous system regulates postural control through precise integration of sensory feedback from visual, vestibular, and proprioceptive systems [20]. Following concussion, chronic postural control impairments were observed when participants were tested with eyes closed, but not with eyes open [4]. In the absence of visual feedback, the central nervous system should increase reliance on vestibular and proprioceptive feedback to maintain postural stability—a process known as sensory reweighting [20, 21]. The resulting postural control impairments observed in eyes closed conditions suggests that relying on proprioceptive and vestibular information after concussion may be insufficient due to impairments in sensory reweighting. However, without systematically assessing each sensory systems' individual contribution to postural control, it is difficult to understand the scope and nature of sensorimotor impairments following concussion [22]. We have developed an experimental paradigm designed to assess sensory reweighting [21, 23, 24]. Participants experience visual, vestibular, and proprioceptive stimulation simultaneously. Each sensory stimulus is applied at a different frequency, so we can quantify the individual contributions of that sensory modality by analyzing postural sway at the stimulus frequency. Standing participants are exposed to four experimental conditions: two with a high-amplitude visual stimulus and two with a low-amplitude visual stimulus, two with vibration (i.e., the proprioceptive stimulus) on and two with vibration off. Thus, in examining the changes in sensory reliance across conditions, we can assess how the central nervous system reweights sensory feedback, providing insight into the mechanisms underlying postural control impairments following concussion. The purpose of this study was to examine sensory reweighting for upright stance across three groups: recent concussion, concussion history, and control. Since previous work observed chronic postural control impairments only in eyes closed conditions [4], we hypothesized participants in both the recent concussion and concussion history groups would display sensory reweighting impairments relative to the control group.

2 Methods

2.1 Study Design

This was a cross-sectional study to examine the effects of concussion on sensory reweighting for upright stance.

2.2 Participants

Potential participants were university undergraduate and graduate students recruited from a single university and included both men and women between ages 18 and

30 years. We recruited participants with a recent concussion (i.e., 2 weeks to 6 months post-injury), participants with a positive concussion history (≥ 1 year post-injury), and participants with no concussion history in a ratio of 1:1:2 (Table 1). Exclusionary criteria for all groups included any self-reported lower extremity injury in the past 6 months;

history of vestibular or ocular dysfunction; currently taking any medications affecting balance; history of any neurological disorders; unstable cardiac or pulmonary disease. The University of Delaware institutional review board approved the study (IRB# 1174642) and the participants provided written informed consent. Furthermore, this study was

Table 1 Demographic information and SCAT-5 data for each group

	Recent concussion	Concussion history	Control	<i>p</i>	Effect size
Demographic information					
Age (years)	21.1 ± 2.7	20.8 ± 1.1	22.2 ± 2.9	0.220	0.061
Height (cm)	175.9 ± 10.4	169.9 ± 9.6	172.0 ± 7.7	0.223	0.061
Weight (kg)	72.7 ± 14.0	67.8 ± 12.9	71.6 ± 10.6	0.560	0.024
Sex (<i>n</i>)					
F	8	7	8	0.109	0.295
M	5	5	18		
Race (<i>n</i>)					
African American	0	2	4	0.164	0.373
American Indian/Alaska Native	0	1	0		
Asian	0	0	4		
Multiple races	0	0	1		
Not reported	0	0	2		
White	13	9	15		
Ethnicity (<i>n</i>)					
Hispanic	0	3	4	0.064	0.295
Not Hispanic	11	9	22		
Not reported	2	0	0		
SCAT-5					
Hospitalized for head injury (<i>n</i>)					
No	13	12	26		
Yes	0	0	0		
Migraine hx (<i>n</i>)					
No	7	11	25	0.002*	0.493
Yes	6	1	1		
LD hx (<i>n</i>)					
No	12	12	26	0.225	0.242
Yes	1	0	0		
ADHD hx (<i>n</i>)					
No	11	10	25	0.343	0.205
Yes	2	2	1		
Depression hx (<i>n</i>)					
No	12	12	24	0.614	0.139
Yes	1	0	2		
Symptoms at baseline (<i>n</i>)	4.8 ± 4.7	1.4 ± 2.1	2.9 ± 3.9	0.097	0.093
Symptom severity at baseline	7.5 ± 8.4	1.9 ± 3.0	4.3 ± 6.8	0.111	0.087
Orientation	5.0 ± 0.0	5.0 ± 0.0	5.0 ± 0.0		
Immediate memory	21.6 ± 2.9	22.5 ± 2.5	23.1 ± 3.8	0.443	0.033
Concentration	4.9 ± .4	4.4 ± .7	4.5 ± .9	0.302	0.049
BESS	3.8 ± 3.5	5.2 ± 5.9	3.7 ± 3.5	0.608	0.023
Delayed recall	7.9 ± 1.4	8.5 ± 1.3	8.2 ± 1.5	0.598	0.021

F female, *M* male, *hx* history, *LD* learning disability, *ADHD* attention-deficit/hyperactivity disorder

performed in accordance with the standards of ethics outlined in the Declaration of Helsinki.

2.3 Concussion History and Sport Concussion Assessment Tool: 5th Edition

All participants self-reported concussion history through a standardized form [15, 25]. Concussion was defined as “a change in brain function following a force to the head, which may be accompanied by temporary loss of consciousness, but is identified in awake individuals with measures of neurologic and cognitive dysfunction [26]. Common concussion symptoms include: headache, feeling slowed down, difficulty concentrating or focusing, dizziness, balance problems, or loss of balance.” It was also noted that a concussion can occur without being “knocked out” or unconscious, and that getting your “bell rung” and “clearing the cobwebs” is a concussion. Participants reported sport- and non-sport-related concussions, diagnosed and undiagnosed concussions, the approximate date of injury, age at time of injury, if they lost consciousness/had difficulty remembering before or after the injury, and how many days they experienced symptoms related to the injury.

All participants completed the Sport Concussion Assessment Tool—5th Edition (SCAT-5). Participants self-reported if they had ever been hospitalized for a head injury, diagnosed/treated for headache disorder or migraines, diagnosed with a learning disability or dyslexia, diagnosed with attention-deficit hyperactivity disorder (ADHD), or diagnosed with depression, anxiety, or other psychiatric disorder, and completed a symptom evaluation, cognitive screening (i.e., orientation, immediate memory, concentration, and delayed recall), and a balance examination [i.e., the Balance Error Scoring System (BESS)].

2.4 Sensory Reweighting Paradigm

We have described the sensory reweighting paradigm in detail elsewhere [21, 23, 24], and used it to examine the effects of repetitive head impacts on sensory reweighting for upright stance in soccer players [23, 24]. Briefly, participants stood in the Immersive Labs (Bertec Corporation, Columbus, OH, USA) and experienced simultaneous perturbations to visual, vestibular, and proprioceptive systems. The visual stimulus was a sinusoidal translation of the visual scene at 0.2 Hz, the vestibular stimulus was a sinusoidal ± 1 mA binaural monopolar galvanic vestibular stimulation (GVS) at 0.36 Hz, and the proprioceptive stimulus was bilateral Achilles’ tendon vibration (on, off) at 0.28 Hz. The visual stimulus was presented at two different amplitudes (low vision = 0.2 m, high vision = 0.8 m) to measure the change in gain to vision, an intra-modal effect; and change in gain to GVS and to vibration, both inter-modal effects. There were

also two levels of vibration (on, off) to measure the change in gain to vision and to GVS, both inter-modal effects. Thus, there were four experimental conditions: (1) low vision—with vibration—with GVS; (2) low vision—without vibration—with GVS; (3) high vision—with vibration—with GVS; (4) high vision—without vibration—with GVS. Participants completed 5 135 s trials of each condition for a total of 20 trials. We randomized trial order for each participant. During analysis, we removed 5 s from the beginning and from the end of each trial to allow the sensory perturbations to ramp up and ramp down. We recorded kinematics in QTM (Qualisys Inc., Göteborg, Sweden) at 100 Hz. Twelve reflective markers were placed bilaterally on the temple, acromioclavicular joint, greater trochanter, lateral femoral condyle, lateral malleolus, and first metatarsal head. We used the midpoint of the right and left greater trochanter markers to estimate the COM, which we used for all analyses.

2.5 Data Analysis

We computed the cross-spectral density (CSD) of the COM postural displacement and each sensory modality, as well as the power spectral density (PSD) of each sensory modality using Welch’s method with a 50 s Hanning window and 50% overlap—and averaged across trials. It is important to note that 25 s is an integer multiple of the periods of all three perturbation signals, so the 50-s window contained an integer number of cycles of each perturbation signal. For each modality, we computed the frequency response function at the stimulus frequency (i.e., 0.2 Hz for vision, 0.36 Hz for GVS, and 0.28 Hz for vibration) by dividing the CSD by the PSD resulting in a complex-valued frequency response function. We defined gain as the absolute value of the frequency response function at the stimulus frequency. Since each modality was presented at a different frequency, the gain represented the separate contribution of each modality to a participants’ sway. We defined phase as the angle of the FRF at the stimulus frequency and converted to degrees; thus, phase was a measure of the temporal relationship between postural sway and stimulus motion. Postural sway either led the stimulus (positive values) or lagged behind it (negative values). To calculate COM residual power, we removed the postural sway response at the stimulus frequencies (0.2, 0.28, and 0.36 Hz) by subtracting sinusoids corresponding to the COM Fourier transform at each stimulus frequency, and then calculated the integrated area of the power spectrum [27, 28]. We computed the COM 95% area and sway velocity as described by Prieto and colleagues [29]. We included COM residual power, COM 95% area, and COM sway velocity because spontaneous sway has been shown to naturally fall into one of three groups of highly correlated measures (i.e., time domain distance/area measures, mean velocity, and frequency domain measures) [29]. All analyses

were performed in MATLAB R2020a (MathWorks, Natick, MA, USA).

2.6 Statistical Analysis

There were three groups in the study: recent concussion, concussion history, and control. We compared demographic information SCAT-5 data between the groups using independent samples *t* tests for continuous variables and Chi-squared for count variables. To compare sensory reweighting between groups, we analyzed data using a mixed-effects analysis of variance (ANOVA). The dependent variables included gains and phases to each modality (i.e., GVS, vision, and vibration), COM residual power, COM 95% area, and COM sway velocity. Conditions (vision, 2 levels; vibration, 2 levels) were the within-subjects (repeated) effects. Group was the between-subjects effect. When appropriate, we used post hoc pairwise comparisons to determine group differences. Gains, COM residual power, COM 95% area, and COM sway velocity were not normally distributed, so data were log-transformed before comparison. Since there were nine outcome measures, we defined significance a priori at $p < 0.01$. For all comparisons, we report partial η^2 as a measure of effect size, whereby 0.01 is small, 0.09 is medium, and 0.25 is large. All analyses were performed in IBM SPSS Statistics 27 (Armonk, NY, USA).

Groups differed in migraine history, whereby a higher proportion of participants in the recent concussion group reported being diagnosed/treated for headache disorder or migraines. Thus, we added an analysis comparing those self-reporting a headache disorder or migraines to those without within the recent concussion group. We analyzed data using a mixed-effects ANOVA. The dependent variables included log-transformed gains and phases to each modality, log-transformed COM residual power, log-transformed COM 95% area, and log-transformed COM sway velocity. Conditions (vision, 2 levels; vibration, 2 levels) were the within-subjects (repeated) effects. Group (migraine history; no migraine history) was the between-subjects effect. We defined significance a priori at $p < 0.01$.

3 Results

The recent concussion group included 13 participants with mean $\pm$ standard deviation of 71 ± 62 days following concussion who initiated a return-to-play progression by the time of testing (Table 1). The concussion history group included 12 participants (Table 1), including 7 with a history of 1 concussion, 4 with a history of 2 concussions, and 1 with a history of 4 concussions; all participants were more than 1 year out from their most recent concussion. The control group included 26 participants with no concussion history

(Table 1). Figure 1 provides a visual representation of the means and 95% confidence intervals for the three groups across four conditions for (a) gain to GVS, (b) gain to vision, (c) gain to vibration, (d) GVS phase, (e) vision phase, (f) vibration phase, (g) COM residual power, (h) COM 95% area, and (i) COM sway velocity.

3.1 Gains/Phases to Each Modality

Figure 1a illustrates greater reliance on vestibular feedback in the high vision conditions relative to the low vision conditions, as well as overall higher gains to vestibular stimuli, among the recent concussion and the concussion history groups than the control group. Specifically, the vision-by-group interaction was significant with a large effect size (Table 2). Post hoc comparisons revealed that both the recent concussion group (high vision 95% confidence interval 0.089–0.130; low vision 95% confidence interval 0.065–0.102) and the concussion history group (high vision 95% confidence interval 0.062–0.105; low vision 95% confidence interval 0.047–0.085) had higher gains in the high vision conditions relative to the low vision conditions than the control group (high vision 95% confidence interval 0.037–0.067; low vision 95% confidence interval 0.040–0.067). Results revealed the main effect for group was also significant with a large effect size (Table 2). Post hoc comparisons for the group effect revealed that both the recent concussion group (95% confidence interval 0.078–0.115, $p = 0.001$) and the concussion history group (95% confidence interval 0.056–0.094, $p = 0.038$) had higher gains than the control group (95% confidence interval 0.040–0.066). Alternatively, the vision-by-vibration-by-group and the vibration-by-group interactions were not significant (Table 2).

Figure 1b illustrates higher gains to visual stimuli among the recent concussion and the concussion history groups than the control group. Specifically, results revealed the main effect for group was significant with a large effect size (Table 2). Post hoc comparisons for the group effect revealed that both the recent concussion group (95% confidence interval 0.795–1.159, $p = 0.002$) and the concussion history group (95% confidence interval 0.633–1.012, $p = 0.018$) had higher gains than the control group (95% confidence interval 0.494–0.752). Alternatively, the vision-by-vibration-by-group, the vision-by-group, and the vibration-by-group interactions were not significant (Table 2).

Figure 1c illustrates no group differences in gain to vibration (Table 2). Specifically, results revealed the main effect for group and the vision-by-group interaction were not significant. Gain to vibration cannot be calculated when the vibration is turned off, so there is no vibration-by-group interaction comparison. Figures 1d–f illustrates no group differences in sensory modality phases (Table 2).

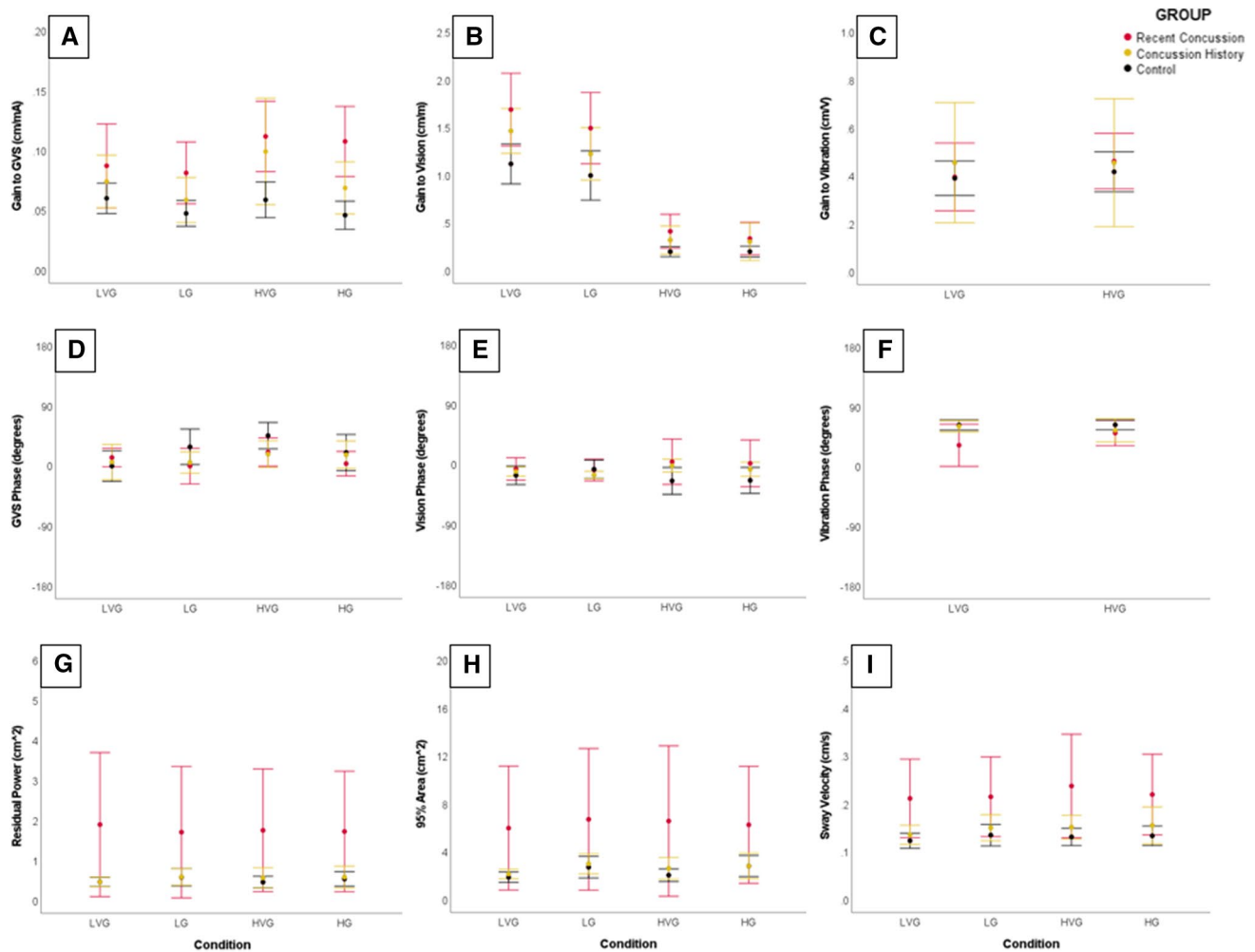


Fig. 1 Means and 95% confidence intervals for the three groups (i.e., recent concussion, concussion history, and control) across four conditions for **a** gain to GVS, **b** gain to vision, **c** gain to vibration, **d** GVS phase, **e** vision phase, **f** vibration phase, **g** COM residual power, **h** COM 95% area, and **i** COM sway velocity. Red represents the recent

concussion group, yellow represents the concussion history group, and black represents the control group. Condition abbreviations: *LVG* low vision—with vibration—with GVS, *LG* low vision—without vibration—with GVS, *HVG* high vision—with vibration—with GVS, *HG* high vision—without vibration—with GVS

3.2 COM Residual Power, 95% Area, and Sway Velocity

Figure 1g illustrates higher COM residual power among the recent concussion group than among both the concussion history and the control groups. Specifically, results revealed the main effect for group was significant with a medium-to-large effect size (Table 2). Post hoc comparisons for the group effect revealed that the recent concussion group (95% confidence interval 1.007–2.507) had higher power than both the concussion history group (95% confidence interval –0.242 to 1.318, $p = 0.042$) and the control group (95% confidence interval –0.032 to 1.028, $p = 0.002$). Alternatively, the

vision-by-vibration-by-group, the vision-by-group, and the vibration-by-group interactions were not significant (Table 2). Figure 1h illustrates no group differences in COM 95% area (Table 2). Figure 1i illustrates higher COM sway velocity among the recent concussion group than among both the concussion history and the control groups. Specifically, results revealed the main effect for group was significant with a large effect size (Table 2). Post hoc comparisons for the group effect revealed that the recent concussion group (95% confidence interval 0.174–0.266) had higher power than both the concussion history group (95% confidence interval 0.099–0.195, $p = 0.044$) and the control group (95% confidence interval 0.097–0.162, $p = 0.001$). Alternatively, the vision-by-vibration-by-group, the vision-by-group, and the vibration-by-group interactions were not significant (Table 2).

Table 2 Results of the mixed-effects ANOVA for gain to each modality across groups (i.e., recent concussion, concussion history, and control)

Source	Gain to GVS			Gain to vision			Gain to vibration		
	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2
Vision $\times$ group	6.905	0.002*	0.223	0.166	0.847	0.007	1.634	0.206	0.064
Vibration $\times$ group	2.204	0.121	0.084	0.919	0.406	0.037			
Vision $\times$ vibration $\times$ group	0.043	0.958	0.002	2.299	0.111	0.087			
Group	7.129	0.002*	0.229	6.553	0.003*	0.214	0.090	0.914	0.004
Source	GVS phase			Vision phase			Vibration phase		
	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2
Vision $\times$ group	0.707	0.498	0.029	2.557	0.088	0.096	2.708	0.077	0.101
Vibration $\times$ group	0.546	0.583	0.022	0.787	0.461	0.032			
Vision $\times$ vibration $\times$ group	1.532	0.226	0.060	0.172	0.842	0.007			
Group	1.274	0.289	0.050	1.299	0.282	0.051	3.543	0.037	0.129
Source	Residual power			95% area			Sway velocity		
	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2	$F_{2,48}$	p	Partial η^2
Vision $\times$ group	0.163	0.850	0.007	0.653	0.525	0.026	0.389	0.680	0.016
Vibration $\times$ group	0.142	0.868	0.006	0.383	0.684	0.016	0.256	0.775	0.011
Vision $\times$ vibration $\times$ group	0.944	0.396	0.038	1.669	0.199	0.065	0.322	0.727	0.013
Group	5.434	0.007*	0.185	3.626	0.034	0.131	6.639	0.003*	0.217

The * indicates statistical significance at adjusted alpha-level, $p < 0.01$

3.3 Diagnosed or Treated for Headache Disorder or Migraines

A higher proportion of participants in the concussion group reported being diagnosed/treated for headache disorder or migraines than in other groups (Table 1). Although sample sizes were relatively small, there were no significant differences between the groups (Table 3; Fig. 2). Nonetheless, this is a potential confounding factor (see Sect. 4.1).

4 Discussion

The purpose of this study was to examine how processing of sensory information for control of upright stance may change following concussion. We hypothesized that participants in both the recent concussion and the concussion history groups would have sensory reweighting impairments relative to the control group, but the results did not support our hypothesis. Instead, participants in both the recent concussion and the concussion history groups had higher gains to visual and vestibular stimuli than participants in the control group. These findings suggest that individuals have abnormal dependence on visual and vestibular feedback in destabilizing environments following concussion, resulting in sensory sensitivity for postural control. There were no differences between the groups in gain to vibration, sensory reweighting (i.e., sensory modality-by-group interactions),

or sensory modality phases. However, participants in the recent concussion group had higher residual power (greater sway across the entire spectrum) and sway velocity than participants in both the concussion history and the control groups. Recovery in conventional sway measures among the concussion history group may indicate that individuals restored function or learned to compensate for less precise sensory feedback through greater reliance on sensory integration to cancel out the noise (i.e., the Bayesian principle).

Although most clinicians and researchers recommend postural control assessments as part of a multimodal concussion examination [7–9, 30, 31], and postural control impairments are both diagnostic and prognostic indicators of concussion [5, 32–34], the physiological mechanisms underlying postural control impairments following concussion remain poorly understood [16–18]. Alterations in sensory, nervous, and/or motor systems can negatively affect postural control. System redundancy allows compensation strategies (e.g., sensory reweighting) to maintain upright stance in the event of alterations in a control system. Current clinical postural control assessments, such as the BESS test, examine the participant's ability to maintain standing balance during challenging sensory conditions, such as with the eyes closed or an unstable surface [35]. However, they do not identify specific components of the system that may be effected because of the concussion. Our sensory reweighting paradigm began to address these limitations by assessing individually, visual, vestibular, and proprioceptive systems,

Table 3 Results of the mixed-effects ANOVA for gain to each modality within the recent concussion group between individuals who self-reported that they were diagnosed or treated for headache disorder

or migraines and those who self-reported no headache disorder or migraine history

Source	Gain to GVS			Gain to vision			Gain to vibration		
	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2
Vision $\times$ migraine	1.481	0.249	0.119	0.332	0.576	0.029	0.693	0.423	0.059
Vibration $\times$ migraine	3.124	0.105	0.221	2.304	0.157	0.173			
Vision $\times$ vibration $\times$ migraine	0.336	0.574	0.030	0.879	0.369	0.074			
Migraine	0.200	0.664	0.018	0.127	0.728	0.011	0.853	0.376	0.072
Source	GVS phase			Vision phase			Vibration phase		
	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2
Vision $\times$ migraine	1.101	0.317	0.091	1.914	0.194	0.148	1.799	0.207	0.141
Vibration $\times$ migraine	3.245	0.099	0.228	0.000	0.990	0.000			
Vision $\times$ vibration $\times$ migraine	3.031	0.110	0.216	0.001	0.981	0.000			
Migraine	0.464	0.510	0.041	3.178	0.102	0.224	0.046	0.835	0.004
Source	Residual power			95% area			Sway velocity		
	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2	$F_{1,11}$	p	Partial η^2
Vision $\times$ migraine	0.757	0.403	0.064	0.173	0.685	0.015	0.008	0.932	0.001
Vibration $\times$ migraine	0.204	0.660	0.018	0.841	0.379	0.071	1.274	0.283	0.104
Vision $\times$ vibration $\times$ migraine	0.059	0.813	0.005	0.134	0.721	0.012	0.251	0.626	0.022
Migraine	0.008	0.932	0.001	0.009	0.914	0.001	0.014	0.907	0.001

The * indicates statistical significance at adjusted alpha-level, $p < 0.01$

as well as sensory reweighting for upright stance [21]. We observed higher gains to visual and vestibular stimuli following concussion, indicating that concussion participants had greater sway relative to the sensory stimuli than control participants. Similar to our findings, Slobounov et al. demonstrated the destabilizing effect of visual field motion following concussion using a moving room paradigm [36–40]. They not only observed residual impairments of concussion at least 30 days post-injury [36, 37], but also observed a dose–response relationship with worse outcomes after the second concussion than after the first [38]. Furthermore, dizziness is the second most commonly reported acute symptom following concussion with 67–77% of athletes self-reporting dizziness [41]. The visual and vestibular impairments we observed may suggest changes in participant’s perception of sensory cues (e.g., noisier feedback), thus resulting in dizziness or feelings of imbalance. Future work should examine the effects of targeted rehabilitation interventions, such as vestibular rehabilitation therapy on visual and vestibular gains following concussion [41–43].

Typically, in the case of noisier or less precise sensory feedback (e.g., vestibular feedback in fall-prone older adults), individuals have lower gains to the less precise sensory system—lower vestibular gains—and higher gains to the more reliable sensory feedback—higher visual gains [44–49]. When a single sensory system is noisy or less

precise, individuals are able to compensate, but not when two or more sensory systems are inaccurate [44–49]. For example, young healthy adults adapt to visual field motion and ignore the destabilizing visual effects; however, visual field motion has a more destabilizing effect on fall-prone older adults with already noisy vestibular feedback [44–49]. Fall-prone older adults are unable to resolve conflicting sources of information, resulting in ambiguity between self-motion and environmental motion. Following concussion, our findings suggest that both visual and vestibular systems have persistent impairments for control of upright stance. Thus, the higher gains, or abnormal dependence on visual and vestibular stimuli, may be a result of ambiguity in self-versus environmental motion [36]. Sensory-challenge balance exercises have been used successfully to reduce abnormally high sensory gains in fall-prone older adults, and may be a future direction for targeted rehabilitation interventions [50].

Although this study was the first to examine sensory reweighting following concussion, others have examined postural control during quiet stance sub-acutely and chronically post-concussion, as well as under various sensory conditions (e.g., BESS test, Sensory Organization Test (SOT), Clinical Test of Sensory Interaction on Balance) and during dynamic tasks (Functional Gait Assessment, Dynamic Gait Index) [4–6, 14]. On the BESS test and even when utilizing

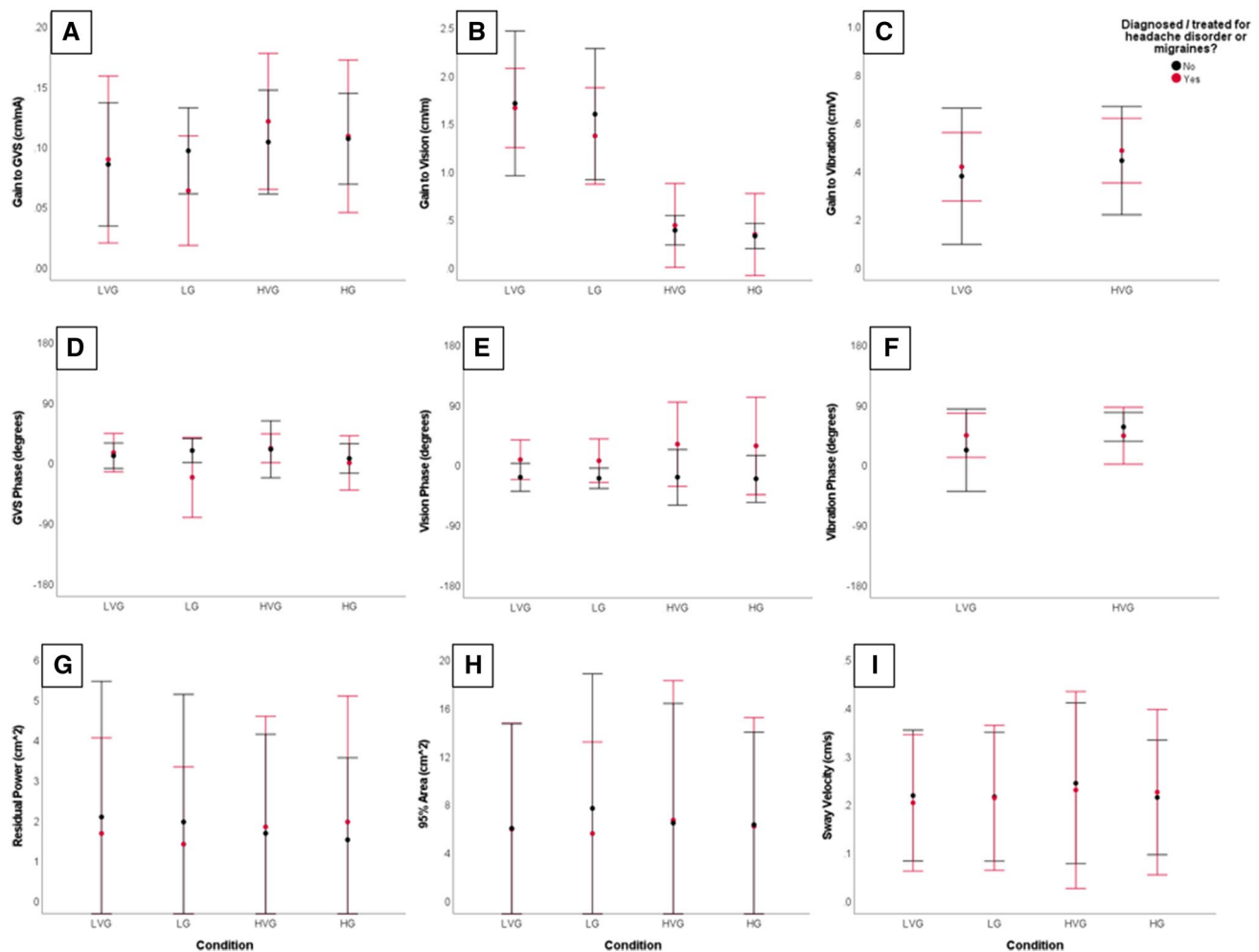


Fig. 2 Means and 95% confidence intervals for the two groups (within recent concussion—diagnosed/treated for headache disorder or migraines, no headache disorder or migraine history) across four conditions for **a** gain to GVS, **b** gain to vision, **c** gain to vibration, **d** GVS phase, **e** vision phase, **f** vibration phase, **g** COM residual power, **h** COM 95% area, and **i** COM sway velocity. Red represents the diag-

nosed/treated for headache disorder or migraines group and black represents the no headache disorder or migraine history group. Condition abbreviations: LVG low vision—with vibration—with GVS, LG low vision—without vibration—with GVS, HVG high vision—with vibration—with GVS, HG high vision—without vibration—with GVS

the instrumented SOT, collegiate student athletes appear to recovery within 2–5 days post-injury [4–6, 14]. However, other approaches, such as dynamic tasks, particularly during dual-task performance suggest persistent postural control impairments at least 30 days post-injury [4–6, 14]. A recent systematic review and meta-analysis suggested there were no persistent deficits in BESS total scores or sway area with the eyes open, but concussed participants had greater sway area than healthy controls with the eyes closed [4]. The authors attributed these findings to impairments in sensory reweighting, but our results clearly show that participants were capable of dynamic reweighting of sensory stimuli. There may be alternate explanations for greater sway area with the eyes closed. For example, participants may have suboptimal sensory weighting or abnormal sensory-to-motor

transformations that inappropriately scale the magnitude of the corrective motor action [18]. Both of these could result in higher gains to sensory stimuli, which is consistent with our findings, and would lead to difficulty with quiet stance balance tasks under challenging visual conditions. These findings have important implications; they can be used in (1) designing rehabilitation interventions, (2) creating more sensitive clinical assessments targeting physiological mechanisms underlying postural control impairments, and/or (3) informing findings regarding the risk of subsequent concussion and/or subsequent musculoskeletal injury.

Contrary to our findings of persistent impairments in visual and vestibular gains among the concussion history group, the concussion history group did not differ from the control group in conventional sway measures. However,

participants in the recent concussion group had higher residual power and sway velocity than participants in both the concussion history and the control groups. Conventional sway measures are related to the quality of sensory feedback, such that noisier or less precise sensory feedback results in greater sway [51]. Consistent with theoretical models and experimental data in older adults with deteriorating sensory feedback [20, 52, 53], individuals can rely on sensorimotor integration to enhance body state estimation and resolve sensory ambiguity—a Bayesian integration process [52, 53]. In other words, sensory integration compensates for increased noise in individual modalities because the combination of two noisy signals is better than either alone [52, 53]. In the concussion history group, conventional sway measures did not differ from the control group, suggesting that these individuals may be relying on sensory integration to cancel out the noise or have simply restored function throughout recovery. Although, the specific recovery curve could not be established with our small sample size and cross-sectional design. Large variability in residual power and in conventional sway measures is likely a result of the variability in time since concussion (range 18–165 days post-injury). It is important to note that all participants initiated a return-to-play progression by the time of testing, so higher residual power and sway velocity among the recent concussion group may indicate that some return-to-play decisions may be premature because residual postural control impairments are still present.

Taken together, our findings suggest there may be alterations in sensorimotor processing in both recent concussion and past concussion history populations in comparison to controls. Furthermore, although some effects persist in past concussion history populations, there are other effects that seem to undergo a degree of recovery or compensation for injuries that happened > 1 year prior to testing. In contrast, there were no differences in BESS scores—a clinical balance assessment—across groups. Participants were young, active adults, and thus, may not have been sufficiently challenged by the BESS test in identifying subtle persistent neurophysiological impairments. Similarly, previous work has proposed the hypothesis that individuals with a previous concussion history can recruit compensatory neural resources to meet task demands [55–58]. Thus, our findings suggest that we may need targeted assessments that can better isolate different systems to probe concussion effects more clearly, which will motivate further investigations to guide clinical care in the future.

4.1 Limitations

This study is not without limitations. Small sample sizes with large variability in time since concussion did not allow for us to examine the recovery trajectory post-injury. Future

work should prospectively examine participants from the time of injury throughout and beyond clinical recovery. We only examined group differences, but the large variability among the recent concussion group may suggest varied etiologies, and individual differences may be important to consider in future work. For example, a high proportion (6/13, 46%) of participants in the recent concussion group reported diagnosis of or treatment for headache disorders or migraine. Within the recent concussion group, we compared those reporting headache disorders or migraine to those without. Although we observed no significant differences between the groups, small sample sizes within groups may have resulted in under-powered comparisons. Therefore, future work must include larger cohorts to examine potential confounding effects of pre-morbidities, such as headache disorder or migraines. Furthermore, without baseline data, it remains unknown if group differences are a result of pre-existing differences or a result of the concussion. Although, considering the “recovery” of the concussion history group relative to the recent concussion group, there are indications these differences may be a result of the concussion. This study recruited a convenient population based on self-reported concussion history, and therefore, there may have been a recall bias in self-reporting and/or a selection bias in participants willing to participate post-injury. Specifically, participants with more severe injury/symptoms acutely post-injury may have been less willing to volunteer and/or participants who suspected they still had lingering impairments post-injury may have been more willing to participate. We only recruited adult participants (ages 18–30 years old), so we cannot extend findings to children and/or older adults, who may experience different outcomes following concussion. Finally, we did not examine any other measures of neurocognitive, neuropsychiatric, or physical function. Therefore, we cannot compare findings to established recovery curves associated with these measures. Future work should include National Institute of Neurological Disorders and Stroke (NINDS) common data elements for sports-related concussion research [54].

5 Conclusion

This study was the first to examine sensorimotor processing during upright stance sub-acutely and chronically following concussion. These findings lend empirical evidence of impairments in visual and vestibular feedback for postural control post-injury, which individuals may learn to compensate for through sensory integration. Considering the increased risk for subsequent concussion and/or subsequent musculoskeletal injury following recovery from concussion, this work has important implications for understanding residual physiological impairments post-concussion.

Future work should replicate these findings in a larger cohort prospectively throughout recovery and should examine the effects of targeted rehabilitation interventions such as vestibular therapy on visual and vestibular gains.

Declarations

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Conflict of interest Fernando V. Santos, as employee of Bertec Corporation, is developing products and has financial interest related to the research described in this paper. All the other authors declare that they have no competing interests.

Ethics approval The University of Delaware institutional review board approved the study.

Consent to participate All the participants provided informed written consent.

Consent for publication Not applicable.

Availability of data and material All data summarized in this publication will be made available upon email request to the study PI and upon completion of a data-sharing agreement.

Code availability Not applicable.

Author contributions JBC, FVS, and FKY carried out the data collection and analysis; JBC drafted the manuscript and performed the statistical analysis; JBC, FVS, FKY, TAB, and JJJ conceived of the study, participated in its design, read and approved the final version of the manuscript, and agree with the order of presentation of the authors.

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